

# Nicholas Kantack

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## Education

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### University of Oklahoma

Bachelor of Science: Engineering Physics  
Graduated May 2016  
Cumulative GPA: 3.98

### University of Virginia

Master of Science: Electrical Engineering  
Graduated December 2020  
Cumulative GPA: 3.87

## Work Experience

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### Amazon Web Services

July 2022 - Present

Software Development Engineer (L5)

- Backend developer for Amazon Connect (Java, Javascript, EC2, Fargate, DynamoDB)
- Principal architect for the rule-based and deep-learning-based algorithms powering Amazon Connect's estimated wait time feature (PyTorch)

### Johns Hopkins Applied Physics Laboratory

July 2018 - July 2022

Software Developer/Electrical Engineer

- Software developer building Android and desktop apps for software defined radios (Java), deep sea embedded devices (C++), and physical simulation (Javascript/Unity)
- Principal investigator for six research grants involving AI explainability in medicine (PyTorch), RL in human-machine teams, XR device applications (Unity/HoloLens), and thin film RF device prototyping (CUDA)

### IntriCon Corporation

May 2016 - July 2018

Electronics Manufacturing Engineer

- Device engineer prototyping sub-millimeter scale inductors for implantable medical devices
- Developer of custom applications for device physics simulation (NodeJS, Python, CUDA)

## Skills

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| <b>Proficient</b> | AWS (ECS, Lambda, Fargate, EC2, Bedrock, IAM, SageMaker, DynamoDB, Connect), AWS CDK, Java, Javascript, Python, PyTorch, Docker, infrastructure as code, probability theory & continuous statistics, statistical tests & data analytics, modeling & simulation, calculus & differential equations, machine learning, neural networks, Android, Tampermonkey, OpenSCAD, Agile, 3D printing, digital circuits, VR/AR |
| <b>Familiar</b>   | C++, C#, Tensorflow, CUDA, R, AutoCAD, Unity, Matlab, LaTeX, NodeJS, AngularJS, SQL, PCB design, RF systems, signal processing, embedded devices, NLP, Deep RL, robotics, AWS Braket, quantum circuits                                                                                                                                                                                                             |

## Publications and Projects

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Visit [nickkantack.github.io](https://nickkantack.github.io) for more information.